Twelve-critical graphs with $\left(\frac{2}{5}+o(1)\right)n^2$ edges

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Abstract

Let $f_k(n)$ denote the maximum number of edges in a k-critical graph on n vertices, where criticality is with respect to edge deletion. We construct a sequence of twelve-critical graphs G_s with $|V(G_s)| \to \infty$ and

$$e(G_s) = \left(\frac{2}{5} + o(1)\right) |V(G_s)|^2.$$

Since $2/5 > 3/8$, this disproves $f_{12}(n) \sim 3n^2/8$, and hence the general asymptotic formula proposed in Erdős Problem 917. The construction combines a coloring module of Pegden with K_5-saturated graphs of sublinear maximum degree.

1 Introduction

All graphs are finite and simple. Following Erdős Problem 917, a graph is k-critical if it has chromatic number k and deleting any edge lowers its chromatic number. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the largest number of edges in a k-critical graph on n vertices. Erdős [3] conjectured that, for fixed $k \geq 6$,

$$f_k(n) \sim \frac{1}{2}\left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2. \tag{1.1}$$

Dirac [2] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [7] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [6, Section 5.1] and Jensen [5]. We show that the coefficient in (1.1) is not optimal for $k = 12$. The problem is recorded as Erdős Problem 917 [4].

Theorem 1. *There is a sequence of twelve-critical graphs G_s such that*

$$|V(G_s)| \longrightarrow \infty, \qquad \frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \frac{2}{5}.$$

In particular, $f_{12}(n) \not\sim 3n^2/8$.

The construction combines the coloring module $U(S,T)$ of Pegden [8] with the saturated graphs of small maximum degree constructed by Alon, Erdős, Holzman, and Krivelevich [1].

2 A coloring module

Let S be an eleven-chromatic graph every proper subgraph of which is ten-colorable, and let T be a ten-chromatic graph every proper subgraph of which is nine-colorable. The module $U(S,T)$ consists of disjoint copies of S and T, together with an independent set

$$A = V(S) \times V(T).$$

Each $(x,y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T, and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x.

The following properties are the $U(11,10)$ case of Pegden's module lemma [8, Lemma 2.5], except that Pegden assumes the factor graphs to be triangle-free. The proof below shows directly that this hypothesis is not needed for the properties used here.

Lemma 2. *Fix a palette of eleven colors.*

1. *In every proper coloring of $U(S,T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ, there is a proper coloring in which the active set uses only α, β, γ, and γ occurs on the active set only at a_0.*
3. *Given an edge e of $U(S,T)$ and distinct colors α, β, there is a proper coloring of $U(S,T) - e$ in which the active set uses only α, β.*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all eleven colors, it contains a vertex colored α and a vertex colored β. Every active vertex in the first row must have color β, so no vertex of T can have color β. The second row similarly excludes α from T. This leaves only nine colors for T, a contradiction.

For (2), color S so that α occurs only at x_0. Color T with the ten colors other than α, so that β occurs only at y_0. Extend these colorings by

$$\operatorname{col}(x,y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0,\ y \neq y_0, \\ \gamma, & (x,y) = (x_0, y_0). \end{cases} \tag{2.1}$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the ten colors other than α, and color all of A with α. If $e \in E(T)$, color $T - e$ with the nine colors other than α, β. Choose $x_0 \in V(S)$ and color S with α only at x_0. Color the row indexed by x_0 with β and all other rows with α.

Suppose instead that e joins an active vertex (x_0, y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0, y_0) to x_0, use (2.1) with the color of (x_0, y_0) changed to α. If it joins (x_0, y_0) to y_0, change that color to β. In either case the only edge that would become monochromatic is the deleted edge. $\square$

3 From saturated graphs to twelve-critical graphs

A graph is K_5-saturated if it contains no K_5, but adding any missing edge creates a K_5. Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a triangle.

Proposition 3. *Let H be a K_5-saturated graph on $v \geq 5$ vertices, with maximum degree $d < v - 1$. Let $h \geq 11$ be odd and suppose*

$$v > 40hd. \tag{3.1}$$

Put $a = 2hv$. There is a twelve-critical graph G with

$$|V(G)| = 5(a + h + 2v) \tag{3.2}$$

and

$$e(G) \geq 10a^2 \left(1 - \frac{d}{v}\right). \tag{3.3}$$

More precisely, if $m = e(H)$, then

$$e(G) = 10(a^2 - 8h^2 m) + 5(2a + 9h + 16v - 79). \tag{3.4}$$

Proof. Construction. Set $\ell = 2v$ and take

$$S = K_8 \vee C_{h-8}, \qquad T = K_7 \vee C_{\ell-7}.$$

Both cycles are odd and have length at least three. Complete graphs and odd cycles have every proper subgraph colorable with one fewer color, and this property is preserved under joins. Indeed, deleting a vertex or an edge within one factor saves a color in that factor; if a joining edge xy is deleted, color each factor so that a prescribed color occurs only at x, respectively y, and identify those two colors. Hence S is eleven-chromatic with every proper subgraph ten-colorable, while T is ten-chromatic with every proper subgraph nine-colorable.

Take five disjoint copies $U_1, \dots, U_5$ of $U(S,T)$, with active sets $A_1, \dots, A_5$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0,1\}$ by any bijection. An active vertex then has the form $(x, (z, \varepsilon))$; give it label z. Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a five-partite graph Z on $A_1 \cup \cdots \cup A_5$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \quad \Longleftrightarrow \quad \pi(u)\pi(w) \in E(H). \tag{3.5}$$

The graph G retains all the edges of the five modules and, between different active sets, takes the complement of Z. There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G.

We record three properties of Z. First, Z is K_5-free: the labeling map sends every clique injectively to a clique of H. Moreover, adding any missing edge between different parts of Z creates a K_5 with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a triangle in $N_H(z) \cap N_H(z')$. Choose vertices with these three labels in the other three parts. Together with u, w, they form the required clique in $Z + uw$. If the labels coincide, say both equal z, choose a vertex z' not adjacent to z; this is possible because $d < v - 1$. The same common-neighborhood triangle is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_4 across any prescribed four parts. Indeed, H has a missing edge; one endpoint together with a triangle in the common neighborhood gives a K_4 in H. Its four labels can be placed in any four parts of Z.

Third, writing $D = \Delta(Z)$, condition (3.1) gives

$$D = 4bd = 8hd, \qquad \ell > 10D. \tag{3.6}$$

The chromatic lower bound. Suppose that G has an eleven-coloring. In each A_i, at least two color classes have size greater than D. Otherwise, choose the unique color with more than D vertices, if there is one, and call it α; if there is none, choose any color. The other ten colors together occupy at most $10D < \ell$ vertices of A_i. Every row, having ℓ vertices, therefore contains a vertex of color α. Each vertex of the copy of S is adjacent to its entire row, so this copy of S would avoid α, contradicting $\chi(S) = 11$.

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Let B_i be the set of colors used more than D times in A_i. The sets $B_1, \ldots, B_5$ are pairwise disjoint and each has size at least two. If, say, $|B_1| \geq 3$, then the other four sets have size exactly two and together $B_1 \cup B_3 \cup B_4 \cup B_5$ contain at least nine colors. Thus A_2 can use at most the two colors in B_2, contrary to Lemma 2(1). Hence $|B_i| = 2$ for every i. Exactly one color remains outside $B_1 \cup \cdots \cup B_5$, and Lemma 2(1) forces it to occur in every active set. Choosing a vertex of this color in each part gives a K_5 in Z, a contradiction. Thus $\chi(G) \geq 12$.

Colorings after edge deletion. Use an eleven-color palette

$$\alpha_1, \beta_1, \ldots, \alpha_5, \beta_5, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G. By the first property of Z, there is a K_5 in $Z + uw$ containing uw, with vertices $a_i \in A_i$. Apply Lemma 2(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i. The only color shared by different active sets is γ. The five vertices of this color are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained an eleven-coloring of $G - e$.

Now let e lie within U_i. Apply Lemma 2(3) to color $U_i - e$, using only α_i, β_i on A_i. Choose a K_4 in Z across the other four parts. In each of those four modules, apply Lemma 2(2) with the selected vertex as the only active vertex of color γ. Once again this is a proper eleven-coloring of $G - e$.

These cases include every edge of G. Recoloring one endpoint of a deleted edge with a twelfth color gives $\chi(G) \leq 12$. Thus $\chi(G) = 12$, and $G - e$ is eleven-colorable for every $e \in E(G)$. Hence G is twelve-critical in the sense of Problem 917.

Moreover, G has no isolated vertices. Therefore every proper subgraph of G is eleven-colorable as well: a proper spanning subgraph is contained in $G - e$ for some missing edge e, while if a vertex x is missing, choose an edge e incident with x; the subgraph is again contained in $G - e$.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each. Thus the number of edges of G between active sets is

$$10(a^2 - 2b^2m) = 10(a^2 - 8h^2m).$$

The two structural graphs have

$$e(S) = 9h - 44, \qquad e(T) = 16v - 35,$$

and each module has $2a$ further edges incident with its active set. This proves (3.4). Finally, $2m \leq vd$ and $a = bv$ give (3.3) from the edges between active sets alone. $\square$

4 Choice of the saturated graphs

We use the following instance of the construction in [1, Section 4, Example 2].

Lemma 4 (Alon–Erdős–Holzman–Krivelevich)**.** *For every prime power $Q \geq 4$, there is a K_5-saturated graph H_Q with*

$$|V(H_Q)| = 13Q^2 + 12Q, \qquad \Delta(H_Q) = 22Q - 3. \tag{4.1}$$

Proof. In Example 2 of [1], set $k = 5$ and give each of the Q^2 additional independent sets one vertex. The core has $12Q(Q+1)$ vertices. A core vertex has $3(3Q-1)$ neighbors in its level, $12Q$ in the other levels, and Q among the additional vertices, so its degree is $22Q - 3$. An additional vertex has degree $12(Q+1)$, which is smaller for $Q \geq 4$. Thus (4.1) holds, and the cited construction is K_5-saturated. □

Proof of Theorem 1. For each integer $s \geq 4$, set

$$h = 3^s, \qquad Q = h^2,$$

and take $H = H_Q$ from Lemma 4. Write $v = |V(H)|$ and $d = \Delta(H)$. Then

$$v = 13h^4 + 12h^2, \qquad d = 22h^2 - 3 < v - 1.$$

The integer h is odd and at least 81. Moreover,

$$40hd < 880h^3 < 13h^4 \leq v,$$

since $13h \geq 13 \cdot 81 > 880$. Proposition 3 therefore gives a twelve-critical graph G_s. Put $a = 2hv$ and $N_s = |V(G_s)|$. We have

$$\frac{N_s}{5a} = 1 + \frac{1}{h} + \frac{1}{2v} \longrightarrow 1, \qquad \frac{d}{v} \longrightarrow 0. \tag{4.2}$$

There are at most $10a^2$ edges between active sets, and the five modules contain $5(2a+9h+16v-79) = O(a)$ edges. Together with (3.3), this gives

$$10a^2\left(1 - \frac{d}{v}\right) \leq e(G_s) \leq 10a^2 + O(a).$$

Dividing by N_s^2 and using (4.2) yields

$$\frac{e(G_s)}{N_s^2} \longrightarrow \frac{10}{25} = \frac{2}{5}.$$

Finally, $N_s \to \infty$ and $f_{12}(N_s) \geq e(G_s)$. Since $2/5 > 3/8$, this subsequence rules out $f_{12}(n) \sim 3n^2/8$. □

Use of generative AI

GPT-6 Astra was used to generate the mathematical proofs and draft the manuscript. GPT-5.6 Sol and Claude Opus 5 were used only for editorial review of the exposition. GPT-6 Astra was run in a research environment containing earlier results produced by GPT-5.6 Sol and Claude Opus 5, but those earlier results did not contribute to the final mathematical arguments. The author reviewed the final manuscript and takes full responsibility for its content.

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